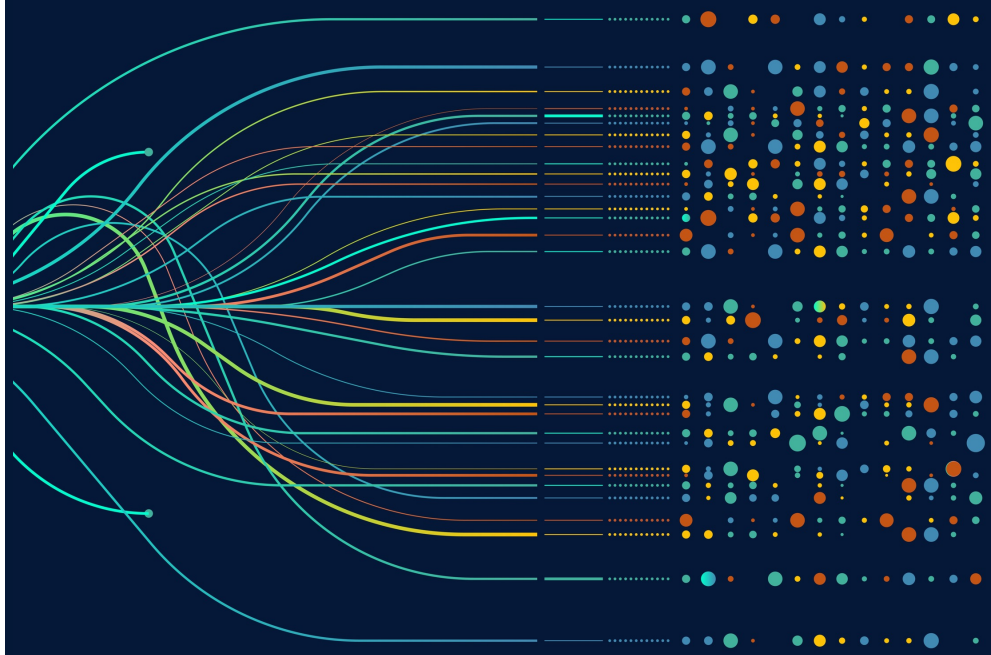
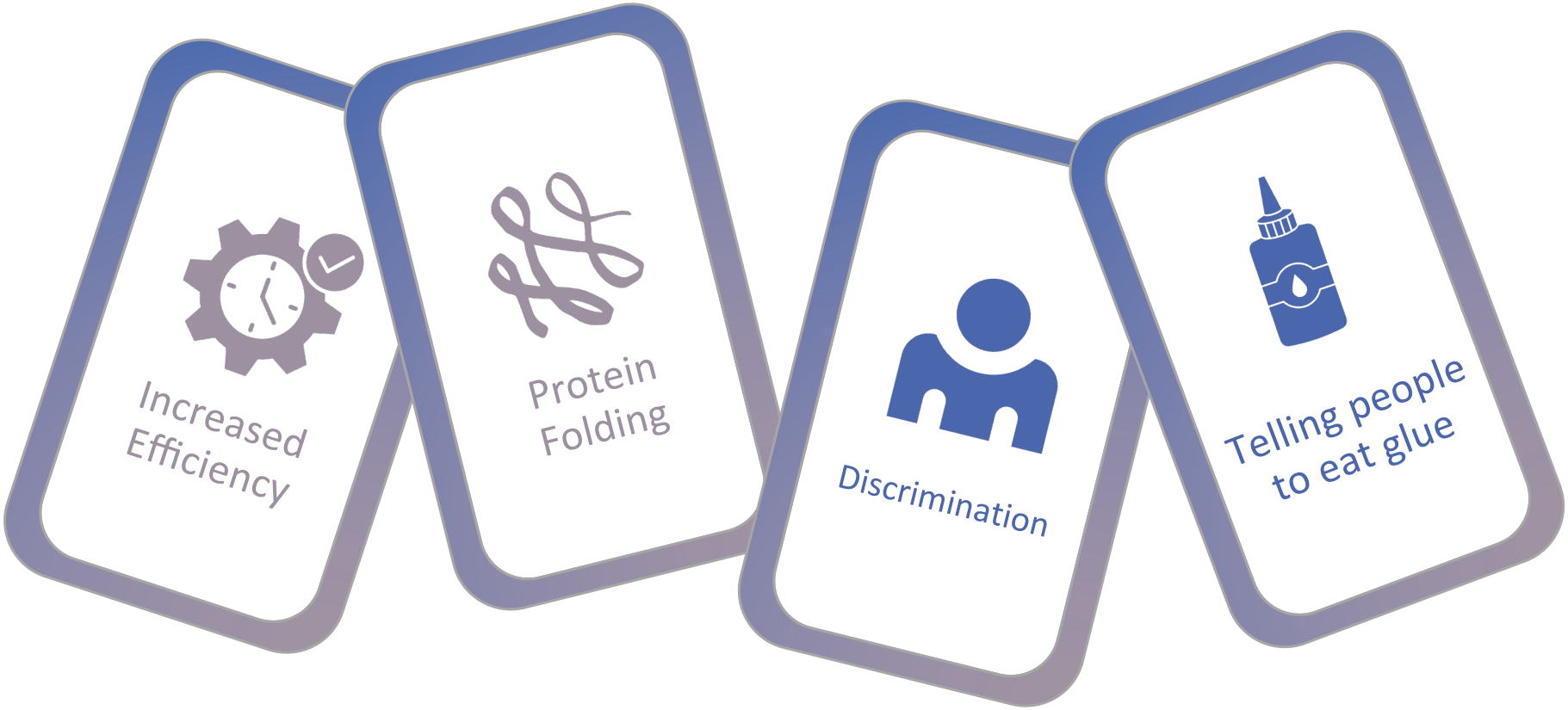


Human-Centered Data Collection for Machine Learning: Lessons from Survey Research

Stephanie Eckman
December 16, 2025

Benefit and Harm from AI



Training Data Collection

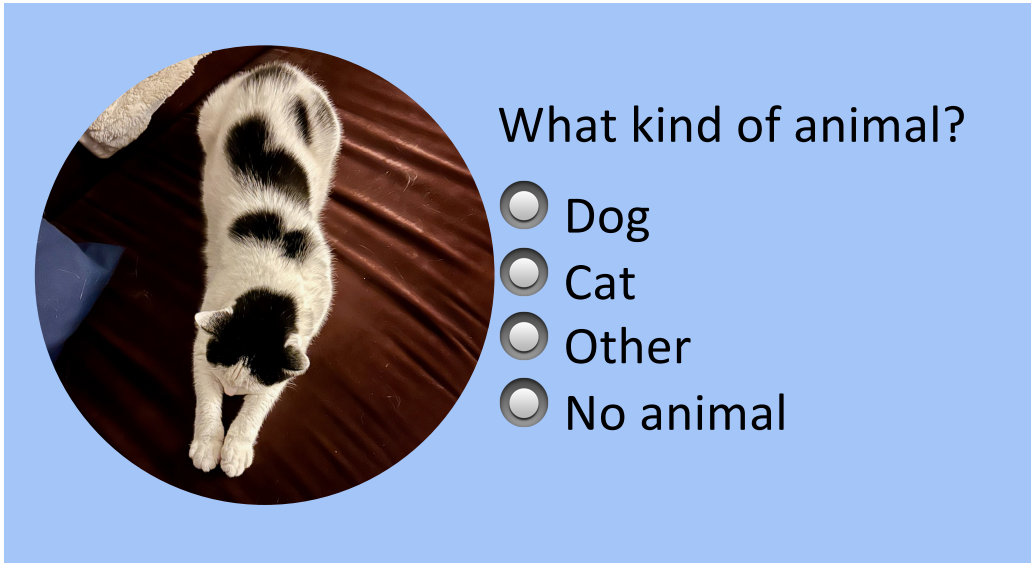
Tend to be objective tasks

Crowdworkers

- Mechanical Turk
- Appen
- Scale AI

Experts

Researchers / Graduate Students



Human Feedback Data Collection

Trained on everything

Human feedback data

- Subjective, opinion tasks
- Collected from crowdworkers

Choose the best response to this query:

What's the weather like?

- It is 28°C and humid.
- It's a warm, humid 28 degrees Celsius out there today – feels pretty sticky!

Core Insight

What kind of animal?

- Dog
- Cat
- Other
- No animal

Choose the best response to this query:

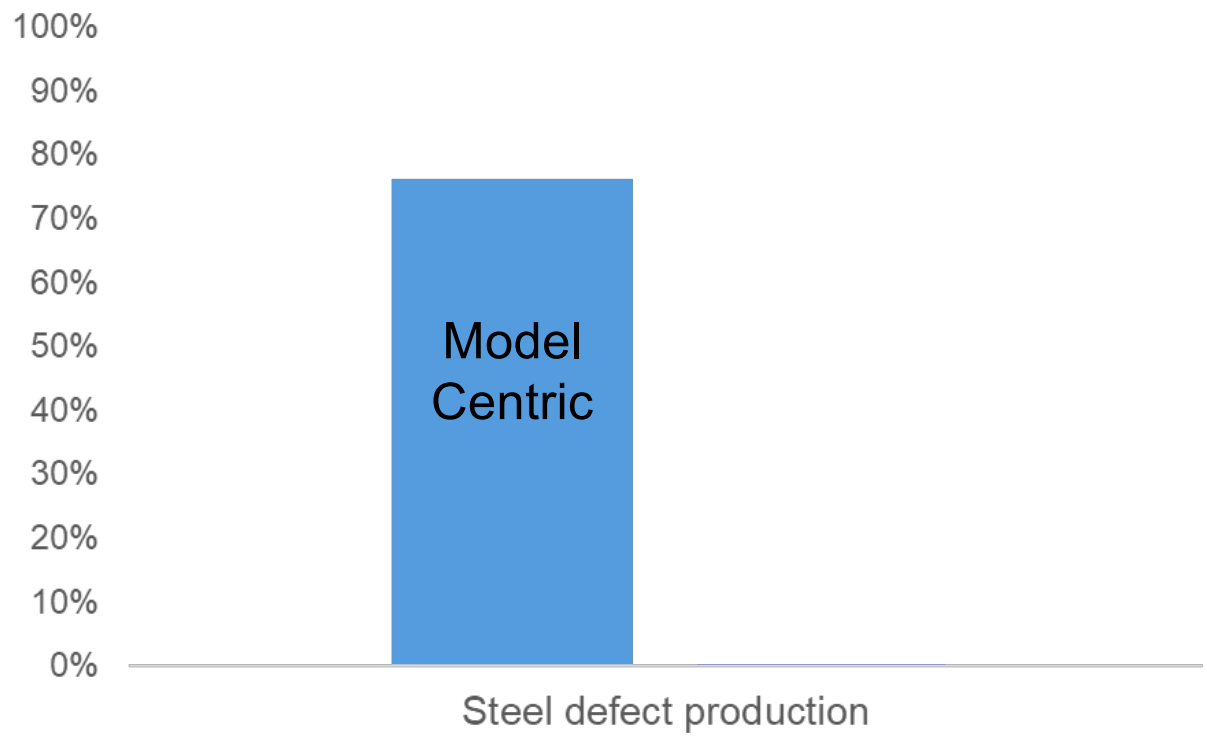
What's the weather like?

- It is 28°C and humid.
- It's a warm, humid 28 degrees Celsius out there today – feels pretty sticky!

A lot like surveys!

Impact of Data Centric AI

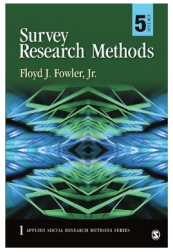
"reasonable algorithm
with good data
is preferable to
a great algorithm with
not-so-good data"
- Andrew Ng

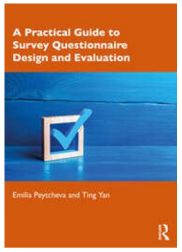


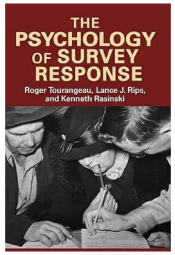
Andrew Ng, Mar 24, 2021 https://youtu.be/06-AZXmwHjo

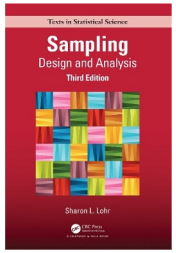
Science of Data Collection, Data Quality

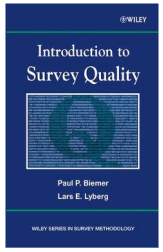
Questions | Sampling | Quality | Journals

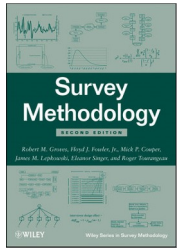


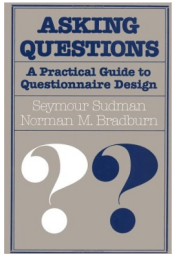


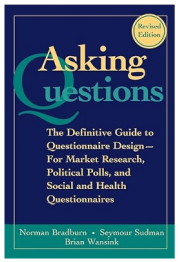


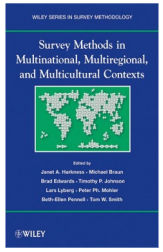


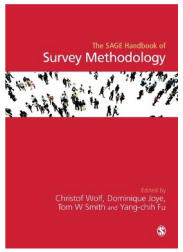


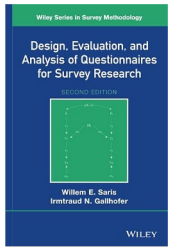


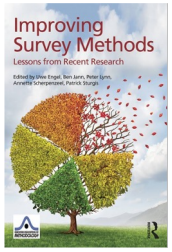


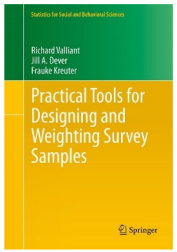


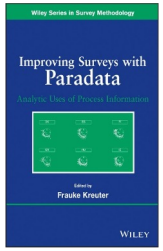


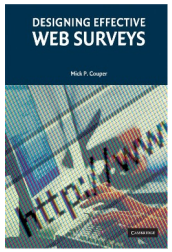












ASA AMERICAN STATISTICAL ASSOCIATION

AAPOR AMERICAN ASSOCIATION FOR PUBLIC OPINION RESEARCH

ESRA EUROPEAN SURVEY RESEARCH ASSOCIATION

WAPOR WORLD ASSOCIATION FOR PUBLIC OPINION RESEARCH

Survey Data Concerns

Training

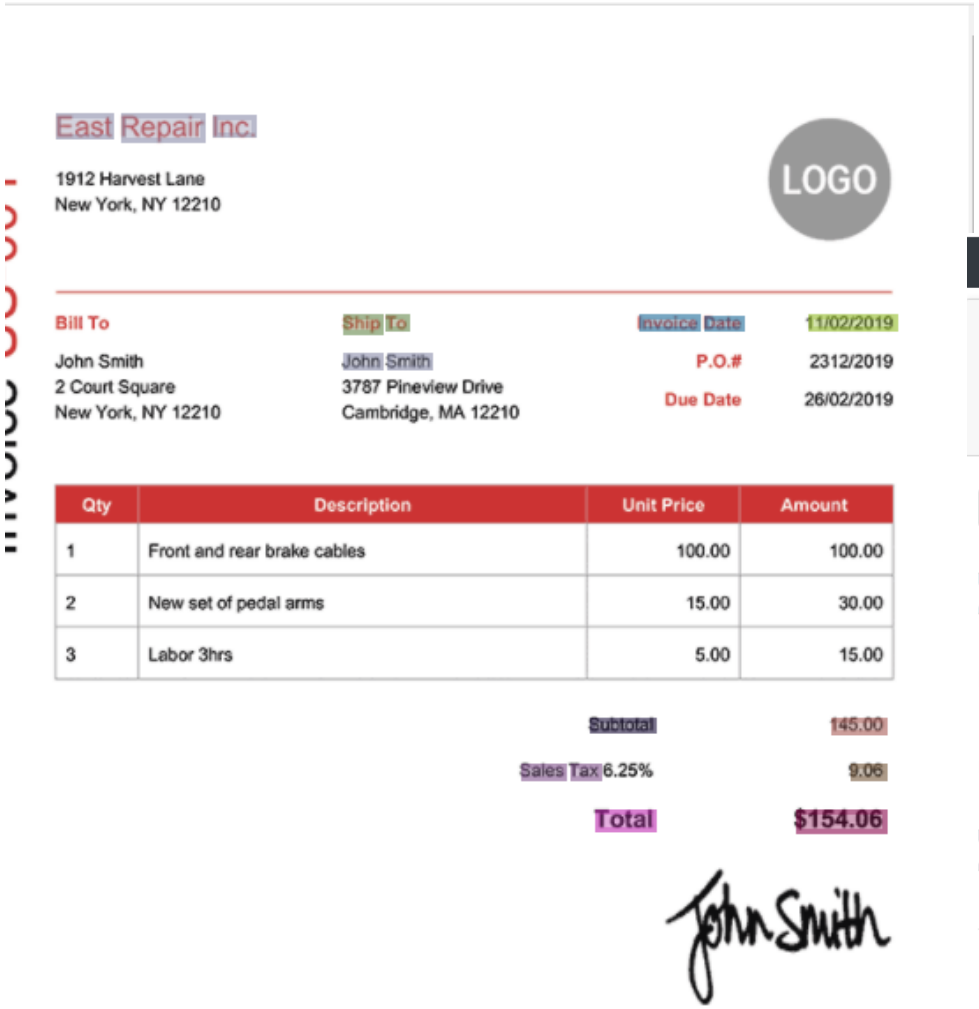
Measurement: Are the answers correct?

labels

Representation: Who responds?

labels

Example Instruments



East Repair Inc.

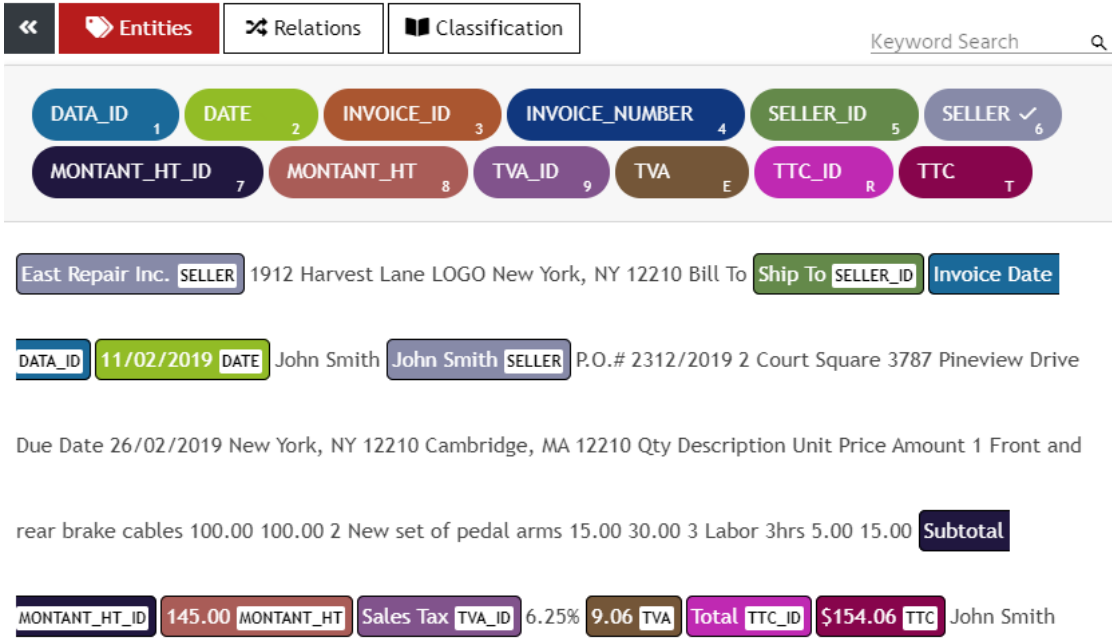
1912 Harvest Lane
New York, NY 12210

LOGO

Bill To	Ship To	Invoice Date	11/02/2019
John Smith 2 Court Square New York, NY 12210	John Smith 3787 Pineview Drive Cambridge, MA 12210	P.O.#	2312/2019
		Due Date	26/02/2019

Qty	Description	Unit Price	Amount
1	Front and rear brake cables	100.00	100.00
2	New set of pedal arms	15.00	30.00
3	Labor 3hrs	5.00	15.00

Subtotal	145.00
Sales Tax 6.25%	9.06
Total	$154.06



https://ubiai.tools/annotate-pdfs-and-images-for-nlp-applications-ubiai/

Search ...

- ○ Bird
- ● Not_Bird
- ○ Unable_to_tell

SUBMIT

https://aclanthology.org/2024.nlperspectives-1.1/

Submit Skip « Page 3 / 11 »

Instruction

Summarize the following news article:

Article text here

Output A

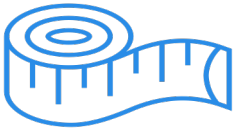
Article summary

Rating (1 = worst, 7 = best)

1 2 3 4 5 6 7

Fails to follow the correct instruction / task ?	Yes	No
Inappropriate for customer assistant ?	Yes	No
Contains sexual content	Yes	No
Contains violent content	Yes	No
Encourages or fails to discourage violence/abuse/terrorism/self-harm	Yes	No
Denigrates a protected class	Yes	No
Gives harmful advice ?	Yes	No
Expresses moral judgment	Yes	No

Ouyang et al. 2022. “Training language models to follow instructions with human feedback”



Measurement:
Are the labels correct?

Finding: Design of Instrument Impacts Data

A

Tweet tweet tweet
Tweet tweet tweet
Tweet tweet tweet

Does this tweet contain hate speech?
Yes
No
Does this tweet contain offensive language?
Yes
No

B

Tweet tweet tweet
Tweet tweet tweet
Tweet tweet tweet

Does this tweet contain hate speech?
Yes
No

Tweet tweet tweet
Tweet tweet tweet
Tweet tweet tweet

Does this tweet contain offensive language?
Yes
No

Version	Offensive Language	Hate Speech
A	52%	27%
B	59%	30%
Corr.	65%	60%

Kern et al. 2023. “Annotation Sensitivity: Training Data Collection Methods Affect Model Performance”

Finding: Instrument Impacts Performance

A

Train

B

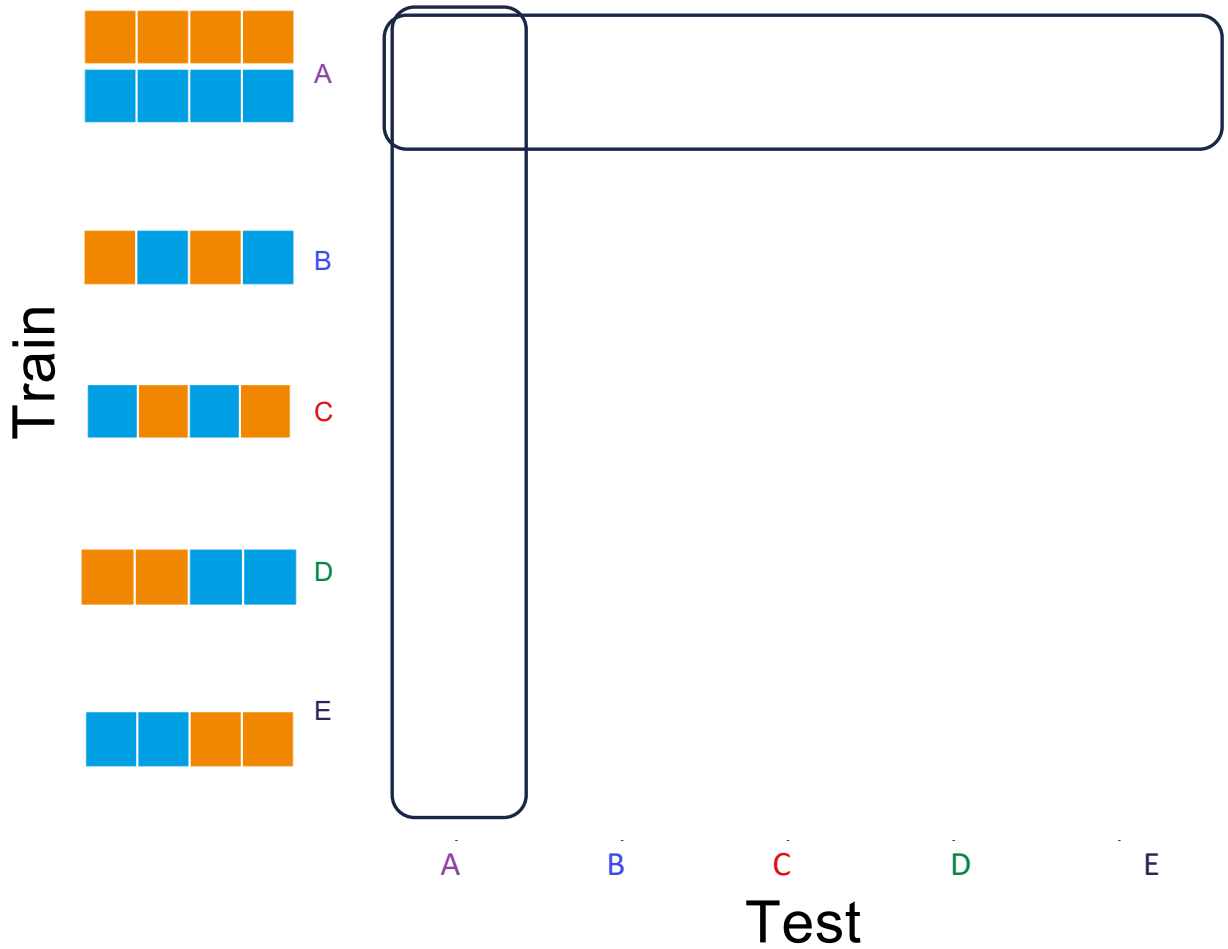
A B

Test

Balanced Accuracy:

Higher is better

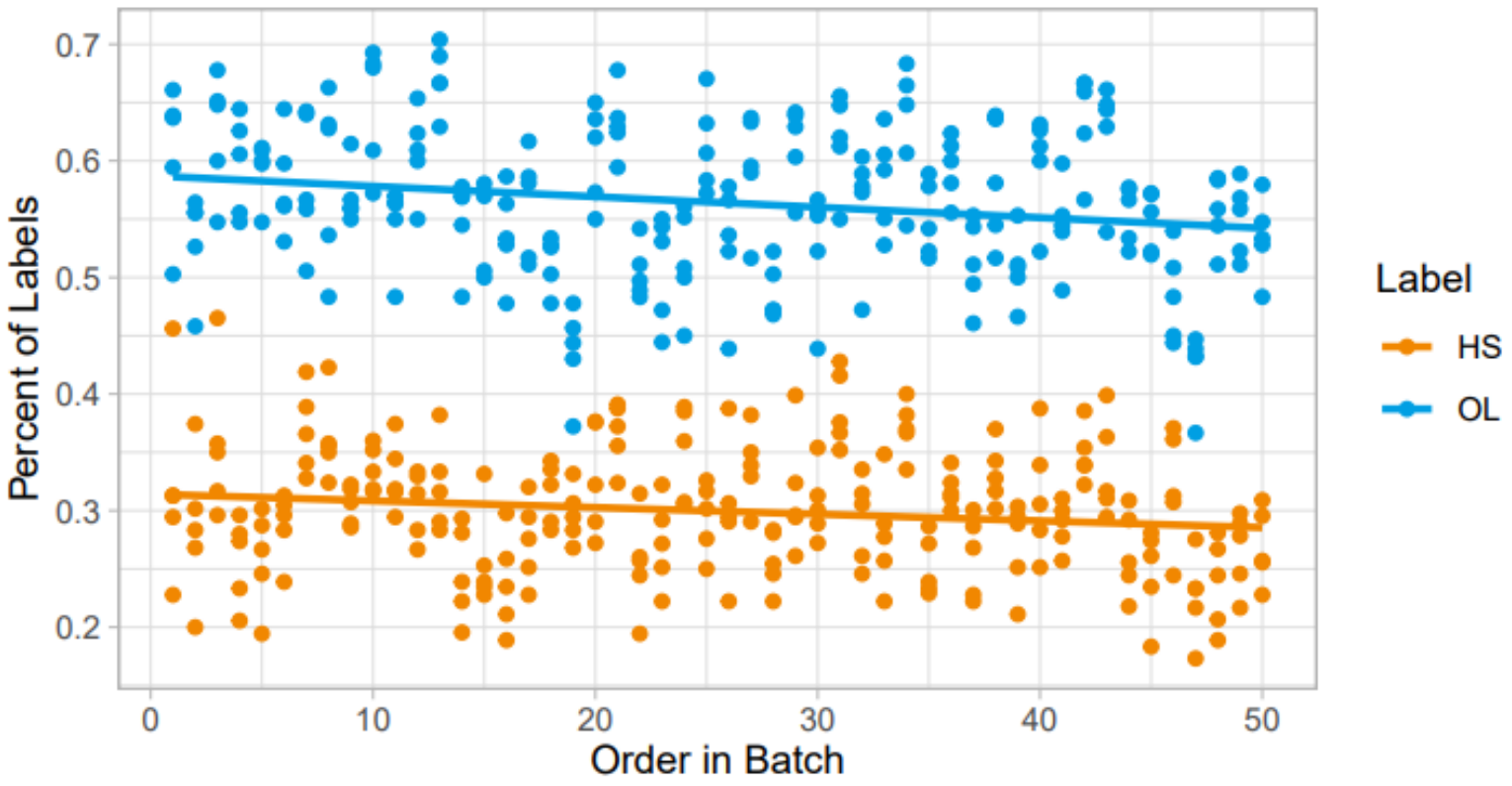
Finding: Instrument Impacts Performance



Balanced Accuracy:

Higher is better

Finding: Order Effects



Beck et al, 2024. https://aclanthology.org/2024.uncertainlp-1.8.pdf

Human-AI Collaboration

Type of Emissions	2019	2020
Total	869	533
Scope 1	0	0
Scope 2	0	0
Scope 3	869	533

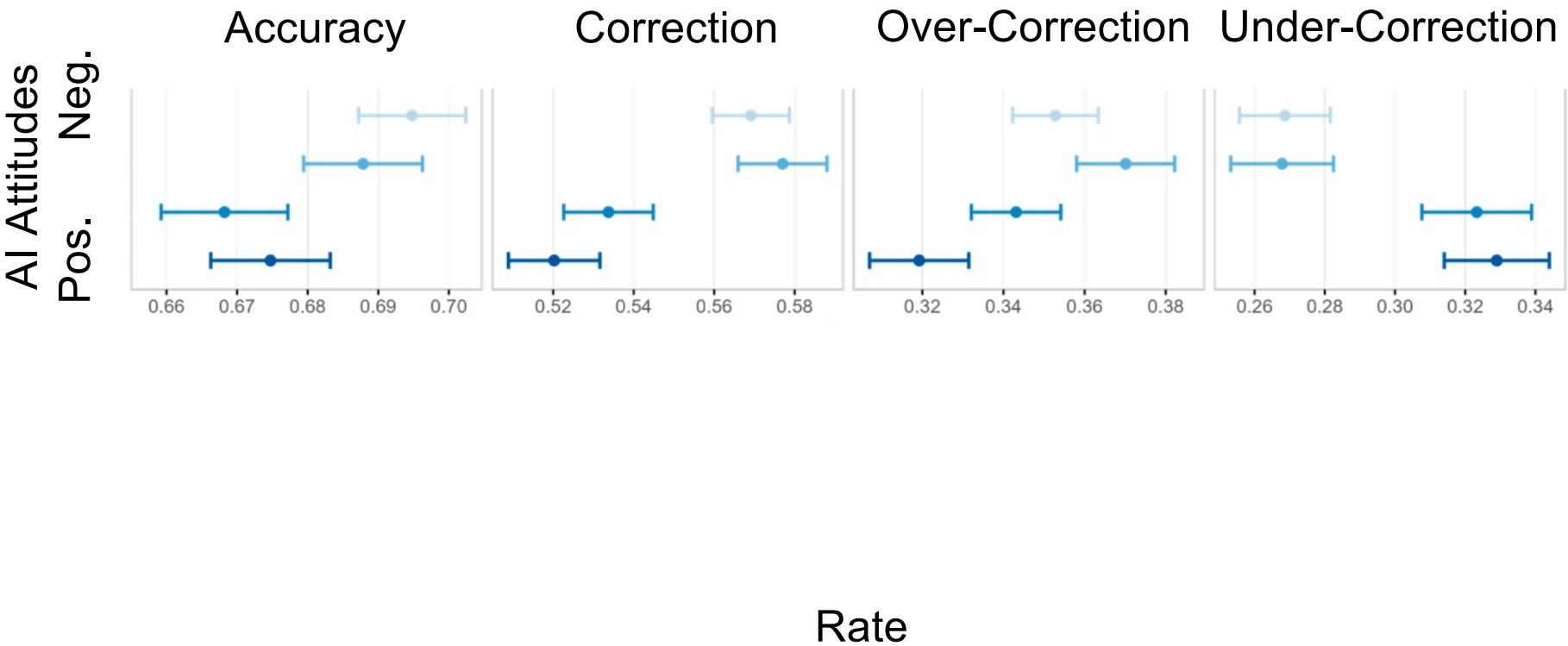
The 2020 Scope 3 emissions are 533, according to the AI

Is this correct?

- Correct
- Incorrect

Beck et al, 2025. Under Review. https://arxiv.org/abs/2509.08514

Finding: Pre-Labeling Can Lead to Biases



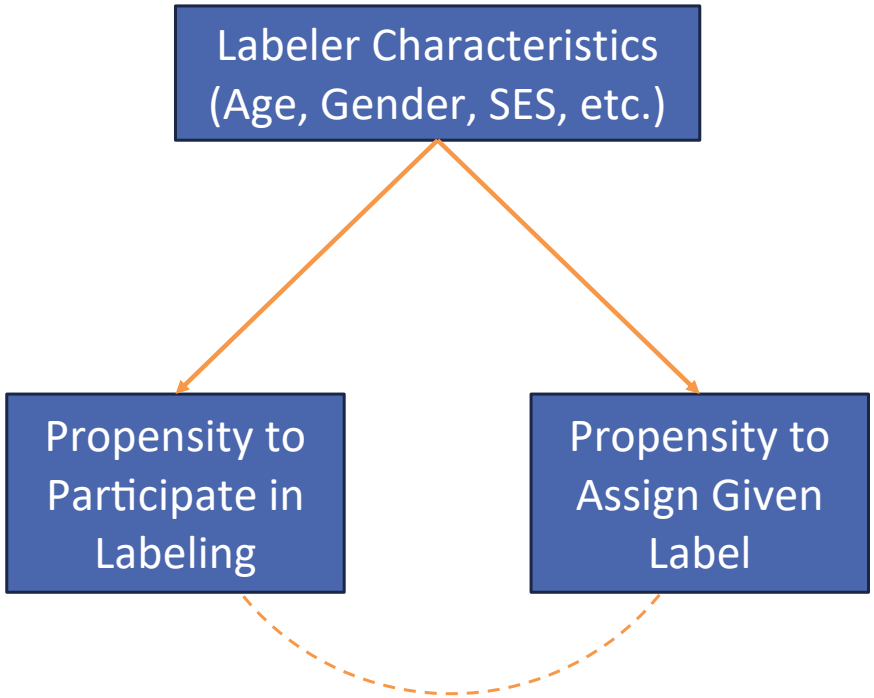
Beck et al, 2025. Under Review. https://arxiv.org/abs/2509.08514

Findings Known in Psychology, Surveys

- Context Effects: Contrast & Assimilation
- Satisficing: Reduce cognitive load
- Motivated Misreporting

Representation: Who labels?

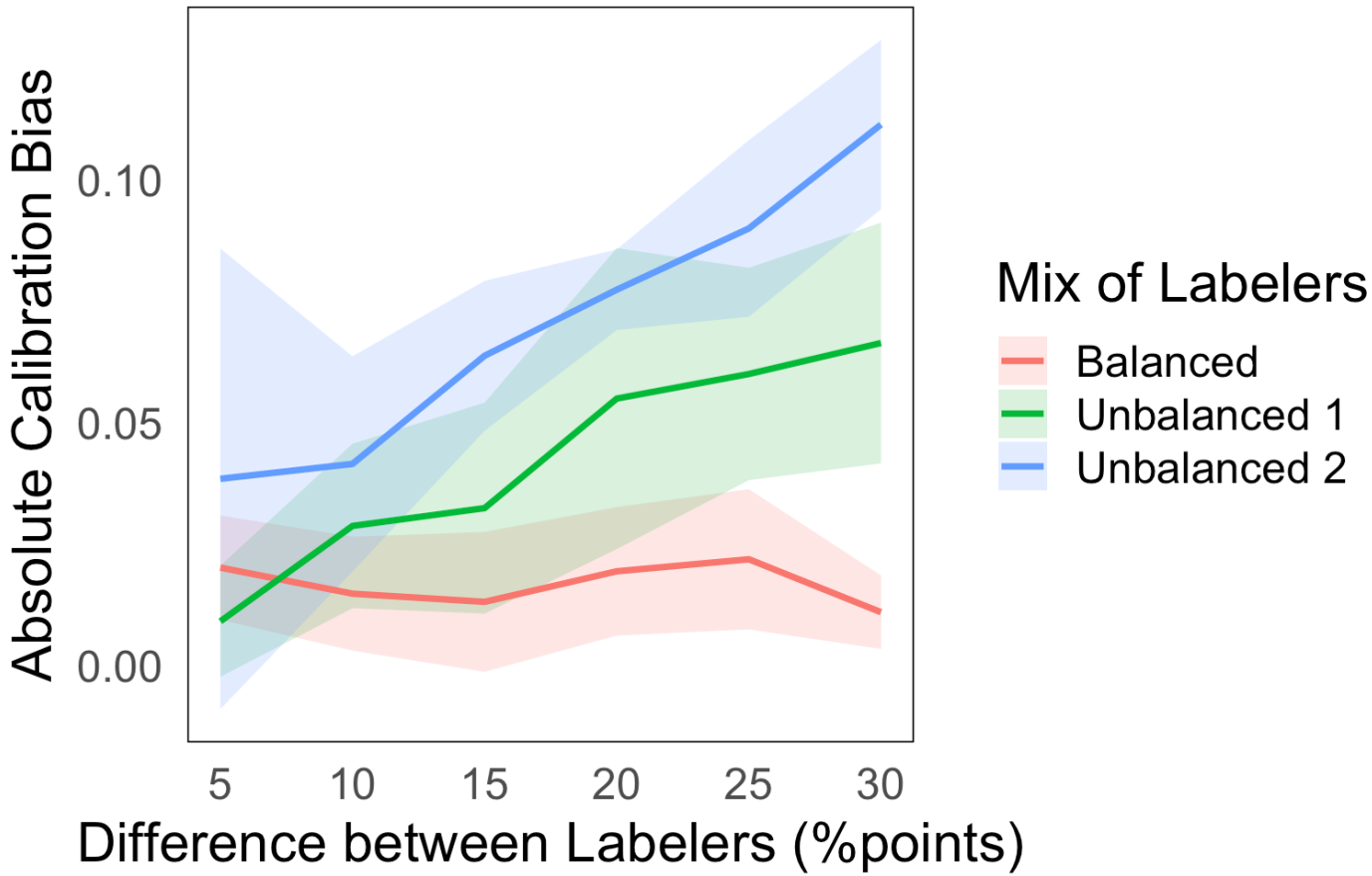
Finding: Who Labels Matters



We could:

- Recruit diverse labelers
- Train labelers to be consistent
- Use weights to adjust training data

Finding: Who Labels Matters



PAIR: Adjust Labels to Match Population

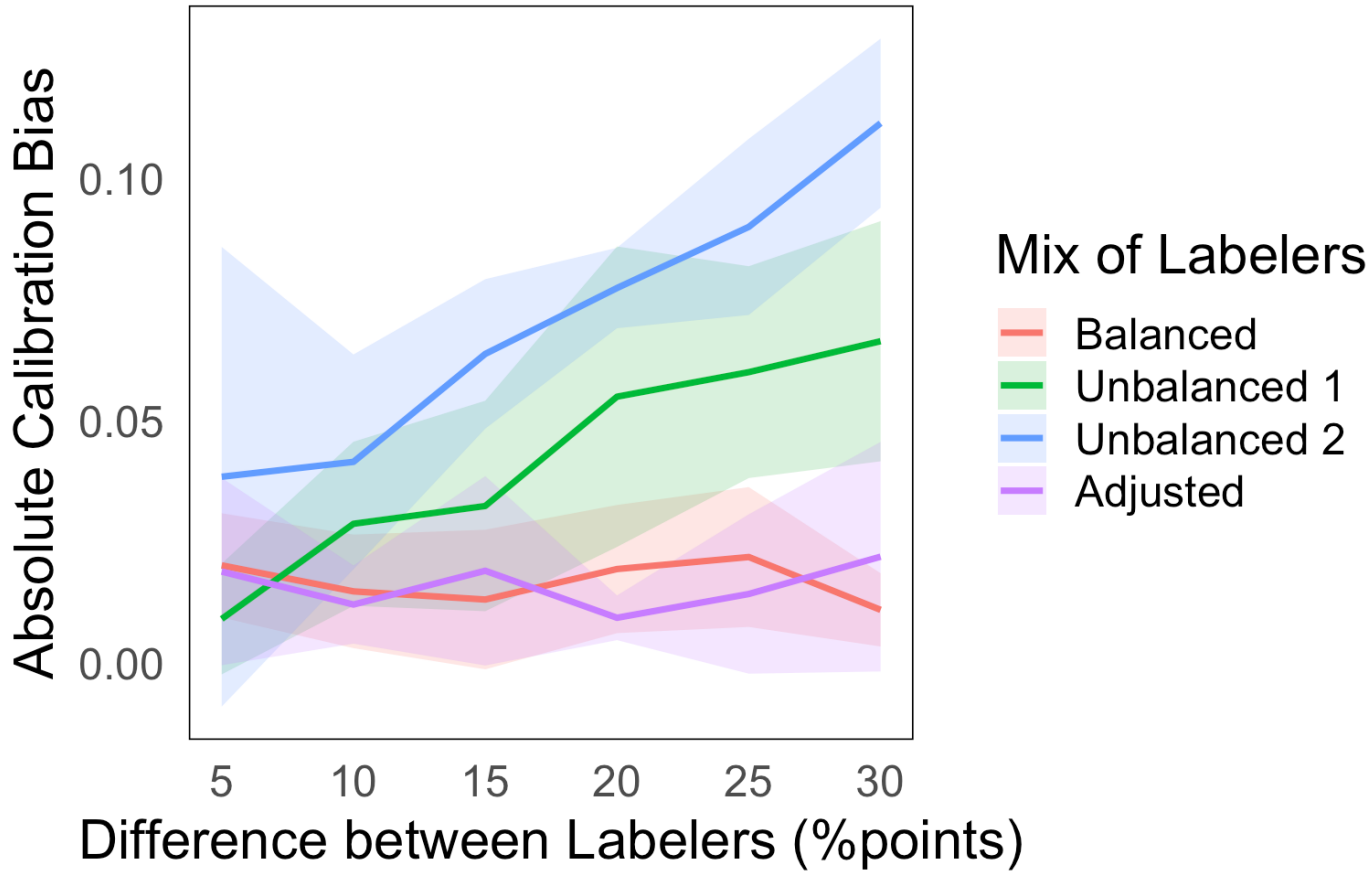
Collect Labeler Characteristics

Calculate Weights

Replicate Instances

Eckman et al. 2024. Position: Insights from Survey Methodology can Improve Training Data for Machine Learning Models

PAIR: Adjust Labels to Match Population



“The bias I am most nervous about is the bias of the human feedback raters“

Sam Altman
March 25 2023 “The Lex Fridman Podcast“

Practical Guidance for Label Collection

Recruit many, diverse labelers

Pay adequately

Simplify instructions: 8th grade reading level

Simplify instrument: Follow survey design practices

Test instructions and instrument

Document everything for replication and transparency

Research Group

Rob Chew
RTI International

Bolei Ma
LMU

Frauke Kreuter
LMU, UMD

Christoph Kern
LMU

Jacob Beck
LMU

Recent Papers

Eckman et al. 2024. Position: Insights from Survey Methodology can Improve Training Data for Machine Learning Models. ICML https://proceedings.mlr.press/v235/eckman24a.html

Kern et al. 2023. Annotation Sensitivity: Training Data Collection Methods Affect Model Performance. EMNLP https://aclanthology.org/2023.findings-emnlp.992/

Beck et al. 2024. Order Effects in Annotation Tasks: Further Evidence of Annotation Sensitivity. UncertaiNLP https://aclanthology.org/2024.uncertainlp-1.8/

Eckman et al. 2025. Aligning NLP Models with Target Population Perspectives using PAIR: Population-Aligned Instance Replication. NLPerspectives https://aclanthology.org/2025.nlperspectives-1.9.pdf

Beck et al. 2025. Bias in the Loop. Working Paper https://arxiv.org/abs/2509.08514

Thank You

Stephanie Eckman
Researcher and Data Scientist
SODA Center, UMD

www.stepheckman.com

AAPOR

AMERICAN ASSOCIATION FOR
PUBLIC OPINION RESEARCH

An LA Love Story of Data, Innovation, and the Quest for Truth

May 13 - 15, 2026
Los Angeles, CA

- Short Course: “Human-Centered Data Collection for Machine Learning: Lessons from Survey Research,” May 12, 2026
- Online Course at Joint Program in Survey Methodology, June 2026

Takeaways / Discussion Questions

Survey scientists can and should consult on training data collection

More research needed to develop best practices

Should probability panels offer training data collection?